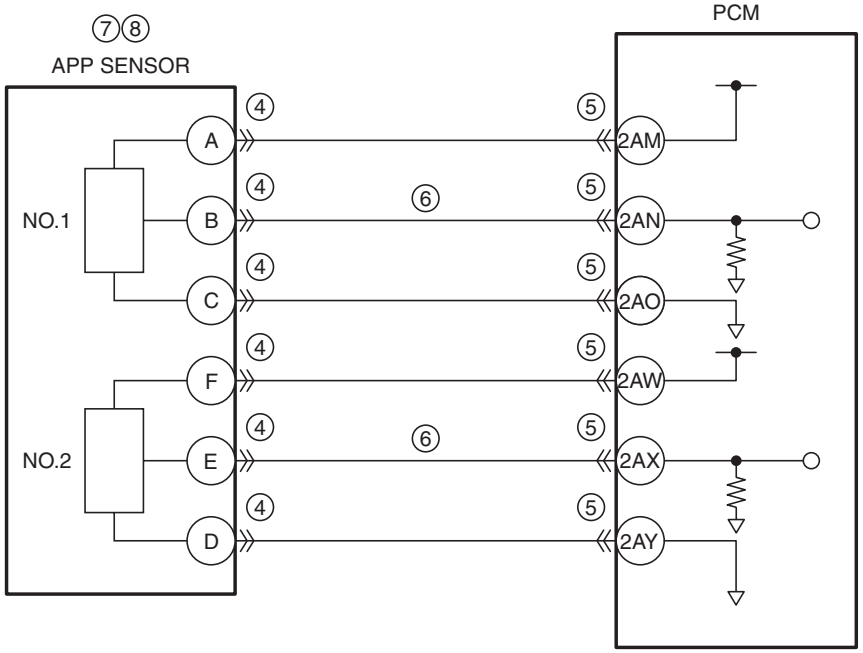
## DTC P2138:00 [SKYACTIV-D 1.5]

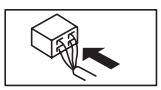
id0102q2708800

| DTC P2138:00 | APP sensor No.1/No.2 correlation problem |
|---|---|
| DETECTION CONDITION | • With the following conditions met, the output voltage of APP sensors No.1 and No.2 is compared, and the difference in the voltage is **0.7 V or more** for a continuous **1 s**.<br>**MONITORING CONDITIONS**<br>— Battery voltage: **8—20 V**<br>— Ignition switched ON (engine off or on)<br>— The following DTCs are not detected:<br>• APP sensor No.1: P2122:00, P2123:00<br>• APP sensor No.2: P2127:00, P2128:00<br>**Diagnostic support note**<br>• This is a continuous monitor (CCM).<br>• The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle.<br>• FREEZE FRAME DATA/Snapshot data is available.<br>• DTC is stored in the PCM memory. |
| FAIL-SAFE FUNCTION | • Limits the engine torque or the upper limit of the engine speed.<br>• Inhibits the auto DPF regeneration control.<br>• Inhibits the EGR control.<br>• Inhibits engine-stop by operating the i-stop function. |
| POSSIBLE CAUSE | • APP sensor connector or terminals malfunction<br>• PCM connector or terminals malfunction<br>• APP sensor No.1 signal circuit and APP sensor No.2 signal circuit are shorted to each other<br>• APP sensor No.1 malfunction<br>• APP sensor No.2 malfunction<br>• PCM malfunction |

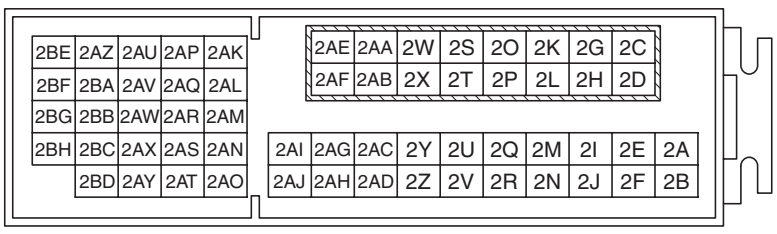


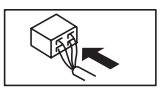

APP SENSOR WIRING HARNESS-SIDE CONNECTOR



PCM WIRING HARNESS-SIDE CONNECTOR



**Diagnostic Procedure**

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 1 | **RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION**<br><br>**Note**<br>• Recording can be facilitated using the screen capture function of the PC.<br><br>• Record the FREEZE FRAME DATA/snapshot data on the repair order. | — | Go to the next step. |
| 2 | **VERIFY RELATED SERVICE INFORMATION AVAILABILITY**<br>• Verify related Service Information availability.<br>• Is any related Service Information available? | Yes | Perform repair or diagnosis according to the available Service Information.<br>• If the vehicle is not repaired, go to the next step. |
| | | No | Go to the next step. |
| 3 | **VERIFY RELATED PENDING CODE AND/OR DTC**<br>• Switch the ignition off, then ON (engine off).<br>• Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• Is the PENDING CODE/DTC P2122:00, P2123:00, P2127:00 or P2128:00 also present? | Yes | Go to the applicable PENDING CODE or DTC inspection.<br>(See DTC P2122:00 [SKYACTIV-D 1.5].)<br>(See DTC P2123:00 [SKYACTIV-D 1.5].)<br>(See DTC P2127:00 [SKYACTIV-D 1.5].)<br>(See DTC P2128:00 [SKYACTIV-D 1.5].) |
| | | No | Go to the next step. |
| 4 | **INSPECT APP SENSOR CONNECTOR CONDITION**<br>• Switch the ignition off.<br>• Disconnect the APP sensor connector.<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals, then go to Step 9. |
| | | No | Go to the next step. |
| 5 | **INSPECT PCM CONNECTOR CONDITION**<br>• Disconnect the PCM connector.<br>• Inspect for poor connection (such as damaged/ pulled-out pins, corrosion).<br>• Is there any malfunction? | Yes | Repair or replace the connector and/or terminals, then go to Step 9. |
| | | No | Go to the next step. |
| 6 | **INSPECT APP SENSOR NO.1 SIGNAL CIRCUIT AND APP SENSOR NO.2 SIGNAL CIRCUIT FOR SHORT TO EACH OTHER**<br>• Verify that the APP sensor and PCM connectors are disconnected.<br>• Inspect for continuity between APP sensor terminals B and E (wiring harness-side).<br>• Is there continuity? | Yes | Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals:<br>• APP sensor terminal B—PCM terminal 2AN<br>• APP sensor terminal E—PCM terminal 2AX<br>**If there is a common connector:**<br>• Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to each other.<br>• Repair or replace the malfunctioning part.<br>**If there is no common connector:**<br>• Repair or replace the wiring harness which has a short to each other.<br>Go to Step 9. |
| | | No | Go to the next step. |
| 7 | **INSPECT APP SENSOR NO.1**<br>• Reconnect all disconnected connectors.<br>• Inspect the APP sensor No.1. (See ACCELERATOR PEDAL POSITION (APP) SENSOR INSPECTION [SKYACTIV-D 1.5].)<br>• Is there any malfunction? | Yes | Replace the accelerator pedal, then go to Step 9.<br>(See ACCELERATOR PEDAL REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) |
| | | No | Go to the next step. |
| 8 | **INSPECT APP SENSOR NO.2**<br>• Inspect the APP sensor No.2. (See ACCELERATOR PEDAL POSITION (APP) SENSOR INSPECTION [SKYACTIV-D 1.5].)<br>• Is there any malfunction? | Yes | Replace the accelerator pedal, then go to the next step.<br>(See ACCELERATOR PEDAL REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) |
| | | No | Go to the next step. |

| STEP | INSPECTION | | ACTION |
|---|---|---|---|
| 9 | **VERIFY DTC TROUBLESHOOTING COMPLETED**<br>• Always reconnect all disconnected connectors.<br>• Clear the DTC from the PCM memory using the M-MDS.<br>(See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].)<br>• Perform the DTC Reading Procedure.<br>(See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)<br>• Is the same DTC present? | Yes | Repeat the inspection from Step 1.<br>• If the malfunction recurs, replace the PCM.<br>(See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)<br>Go to the next step. |
| | | No | Go to the next step. |
| 10 | **VERIFY AFTER REPAIR PROCEDURE**<br>• Perform the “AFTER REPAIR PROCEDURE”.<br>(See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].)<br>• Are any DTCs present? | Yes | Go to the applicable DTC inspection.<br>(See DTC TABLE [SKYACTIV-D 1.5].) |
| | | No | DTC troubleshooting completed. |